



Innovative Applications of O.R.

A mathematical framework to optimize resilience of interdependent critical infrastructure systems under spatially localized attacks

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ABSTRACT

This paper defines the hazards that can cause direct localized damage or interruption as Spatially Localized Attacks (SLAs). The SLAs-induced impact on a critical infrastructure system (CIS) is modeled as the failure of system components that exist within some localized area while those outside this area remain operating. Instead of identifying and analyzing each type of the SLAs-related hazards, this paper makes a worst-case analysis and proposes a mathematical framework to support resilience optimization of interdependent CISs under the worst SLA. For illustrative purposes, this paper mainly considers two types of strategies to enhance CIS resilience, including protecting weak components, and building new components to increase redundancy. The problem is mathematically formulated as a tri-level defender–attacker–defender model, which is exactly solved by a proposed decomposition algorithm. The case study on interdependent power and water systems demonstrates how the proposed approach can not only identify the optimum resilience enhancement strategy as well as the worst-case SLA, but also analyze the importance of considering interdependencies from both the attacker's and the defender's perspectives.

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1. Introduction

Critical infrastructure systems (CISs), such as electric power, water supply, transportation and telecommunication systems, are the backbone for a functioning city or nation as they provide essential services to support the well-being of its citizens. These CISs are not isolated and may be physically, geographically, cyber and logically dependent and interdependent (Ouyang, 2014; Peerenboom, Fisher, Rinaldi, & Kelly, 2002; Rinaldi, Peerenboom, & Kelly, 2001). Strictly speaking, dependencies refer to unidirectional relationships, and interdependencies indicate bidirectional interactions (Rinaldi et al., 2001). Nonetheless, in the literature, dependencies are usually regarded as interdependencies, unless they are particularly referred to, which is also applied in this paper. On the positive side, interdependencies may improve the operational efficiencies of these systems. For example, in China and Japan, geographical interdependencies among some CISs are intentionally designed by constructing utility tunnels, which are passages built underground or aboveground to carry utility lines, such as electricity wires, water pipes, sewer pipes and some communications

utilities. These utility tunnels can facilitate the maintenance of the carried CISs and enhance the land use efficiency.

On the negative side, interdependencies may aggravate the damage consequences (Ouyang, 2014). For example, due to the physical interdependencies, failure in one system can propagate to other systems; due to the geographical interdependencies, some extreme events can break the utility tunnels and make multiple CISs fail simultaneously. These issues have been recognized by many governments and scholars. A lot of countries have initiated critical infrastructure protection plans, such as claiming December as critical infrastructure protection and resilience month by the president of the USA in 2009, releasing critical infrastructure resilience strategy and implementation program in Australia in 2010 (Australian Government, 2010), and publishing critical infrastructure resilience program in UK in 2011 (Secretary of State for Environment, Food and Rural Affairs by Command of Her Majesty, 2011). In these plans, the term resilience was emphasized. In the research field, the number of the resilience-related papers in the Scopus database increased exponentially in the past twenty years (Bergstrom, Winsen, & Henriqson, 2015; Righi et al., 2015). Then, what is resilience?

Despite many definitions for the term resilience, even only focusing on CISs (ASCE Policy Statement, 2013; The Infrastructure Security Partnership, 2011; The White House, 2013), there still exists a consensus. Basically, resilience is the capability of a system to withstand internal/external stresses and recover

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List of Acronyms

CISs	Critical Infrastructure Systems
SLAs	Spatially Localized Attacks
DAD	Defender–Attacker–Defender
ROICISs-SLAs	Resilience Optimization of Interdependent CISs under Spatially Localized Attacks
MIP	Mixed integer programming

from them. To quantify the CIS resilience, various metrics have been proposed based on the system performance curves under disruptions (Cavallaro, Asprone, Latora, Manfredi, & Nicosia, 2014; Franchin and Cavalieri, 2015; Francis & Bekera, 2014; Hosseini, Barker, & Ramirez-Marquez, 2016). Some resilience metric, like reliability and survivability (Ben Yaghlane & Azaiez, 2017), is described by a probability (Chang & Shinozuka, 2004), but most of existing resilience metrics are described as deterministic values. Despite the absence of a common resilience metric, there are several consensuses. First, when considering the resilience, we need to ask ourselves, “resilience to what”, which means that resilience quantification needs to first specify the type of disruptive event. Second, the value of resilience to a disruptive event is mainly affected by system robustness and system recovery rapidity under this event, where the robustness is quantified as system performance level immediately after the event and before any restoration efforts, and the recovery rapidity describes how quickly the system recovers after the event. Specially, in the case of sufficient repair resources, the resilience is mainly affected by the robustness. As an initial step, this paper simply quantifies system resilience as the robustness. In the literature, scholars have studied the robustness or the performance drop immediately after the event for interdependent CISs under different types of disruptive events, including random failures that are usually modeled by randomly removing a certain fraction or number of CIS components (Baxter, Dorogovtsev, Goltsev, & Mendes, 2012; Buldyrev, Shere, & Cwlich, 2011; Gao, Buldyrev, Havlin, & Stanley, 2011; Parshani, Buldyrev, & Havlin, 2010; Rosato, Issacharoff, Tiriticco, & Meloni, 2008; Shao *et al.*, 2011; Svendsen & Wolthusen, 2007; Zio & Sansavini, 2011), natural hazards whose impacts on CIS components are usually modeled according to their fragility curves (Cavallaro *et al.*, 2014; Dueñas-Osorio, James, & Barry, 2007a; Adachi & Ellingwood, 2008; FEMA, 2015; Franchin and Cavalieri, 2015; Hong *et al.*, 2015; Ouyang & Dueñas-Osorio, 2011; Ouyang & Wang, 2015; Poljansek, Bono, & Gutierrez, 2012 & 2017), malicious attacks that are usually modeled as the failure of important components (Dueñas-Osorio, James, Barry, & Ann, 2007b; Huang, Gao, Buldyrev, Havlin, & Stanley, 2011; Nan, Eusgeld, & Kröger, 2013; Zio & Sansavini, 2011; Ouyang, Pan, Hong, & He, 2015; Zio, Golea, & Rocco, 2012a & 2012b; Pinar, Meza, Donde, & Lesieutre, 2010; Wang, Hong, Ouyang, Zhang, & Chen, 2013; Chen, Du, Cao, & Zhou, 2015; Chopra and Khanna, 2015), and some combinations of the above events (Levitin, 2007; Levitin & Hausken, 2009). Actually, the above mentioned disruptive events should strictly be grouped into non-proximity-based events as they do not consider the geographical proximity or correlation of the damaged components.

This paper mainly focuses on spatially localized attacks (SLAs), which are defined as the hazards that can cause direct damage or interruption of system components that exist within some localized area while those outside this area remain operating. For example, on September 11, 2001, the New York terrorist attack caused the full collapse of the WTC1 and the WTC2, and the debris caused the damage of some neighboring buildings. These damaged buildings further caused the damage of many infrastructure components

within 0.21 kilometers from the attack center, including power substations, water pipes, telecommunication centers and subway tunnels. On August 12, 2015, the Tianjin chemical explosion event caused the damage or interruption of some main road segments, a subway station and many other infrastructure components located within 1 kilometers from the attack center. On July 06, 2016, the Wuhan heavy rainstorm caused the whole city as an ocean, and all three railway stations in Wuhan were flooded and interrupted, but the railway interruption was still localized from the whole country view. From the above events, it can be found that the SLAs can be triggered in various ways and some of which may be unexpected and cannot be identified until they occur. Instead of defining and analyzing each of the SLAs-related hazards, this paper makes a worst-case analysis and proposes a mathematical framework to support resilience optimization of interdependent CISs under the worst-case SLA. In the literature, there are some studies on identifying the worst-case localized attack in interdependent CISs. For instance, Patterson and Apostolakis (2007) studied the interdependent CISs in a campus area by dividing the area into a generic hexagonal grid with a small radius 7 meters, and removed the CIS components within each grid to approximate the localized attack for identifying the worst-case attack. Johansson and Hassel (2010) made a similar analysis by considering square attack areas. Nonetheless, the results from these studies depended on the shape of the attack area (square, hexagon, etc.) and how the infrastructure map was partitioned into small attack areas. Recently, Ouyang (2016) and Ouyang, Tian, Wang, Hong, and Mao (2017) modeled the attack area by a circle shaped area and then proposed an exact algorithm to identify the worst-case attack. However, all these studies did not investigate how to mitigate the SLAs for maximizing system resilience under the worst-case SLA, as also not found in other literature by the author.

Despite the reduced number of studies for interdependent CISs under spatially localized attacks, for single CISs and for non-proximity-based malicious attacks, similar problems have been extensively investigated in the literature. This type of problems needs to introduce a virtual attacker who seeks the most harmful attack strategy to disrupt the system and a defender who can take defense measures to minimize the damage consequences. The interactions between the attacker and the defender can be described by a tri-level defender–attacker–defender (DAD) model. The inner level of this model describes how the defender manipulates component flow to minimize the damage consequences, the middle level describes how the attacker disrupts the system to maximize the disruption consequences, and the outer level describes how the defender can optimally protect the system. This modeling framework has been applied to identify the optimal protection strategies for electric power systems (Brown, Carlyle, & Salmeron J & Wood, 2006; Chen, Dong, Hill, & Xue, 2011), a rail system (Alderson, Brown, & Carlyle, 2014), a notional commodity distribution network (Alderson, Brown, & Carlyle, 2015), and some other CISs (Alderson, Brown, Carlyle, & Wood, 2010) under the non-proximity-based malicious attacks. This paper will adapt the framework for interdependent CISs against the spatially localized attacks and address the modeling and solution challenges due to the interdependencies among CISs. For illustrative purposes, this paper mainly considers two types of strategies to enhance system resilience, including protecting weak components, and building new components to increase redundancy. In addition, different from many existing studies in the literature that adopt probabilistic frameworks to model the actions and outcomes of the attacker and defender (Hausken & Levitin, 2009 & 2012; Hausken, 2011; Hausken, 2017), where the defender minimizes the maximal expected damage that an attacker can inflict, this paper uses the worst-scenario approach and the defender minimizes the expected damage given that the attacker selects the worst-case

attack scenario. Hence, the problem in this paper can be described by a deterministic optimization model.

The rest of the paper is organized as follows: [Section 2](#) introduces the model for resilience optimization of interdependent CISs against spatially localized attacks (ROICISs-SLAs). [Section 3](#) provides the solution algorithm. [Section 4](#) shows the results by applying the proposed approach to interdependent power and water systems. [Section 5](#) gives conclusions and future work.

2. Model for resilience optimization of interdependent CISs under spatially localized attacks

Interdependent CISs can be modeled by different approaches, such as agent based approaches, system dynamics based approaches, economic theory based approaches, network based approaches and others ([Ouyang, 2014](#)). As studying spatially localized attacks needs system topological and geographical information, this paper uses a network-based approach, where each CIS is modeled as a network and their interdependencies are modeled by inter-links. In addition, as mentioned in the Introduction section, the resilience of a system under a disruptive event is primarily affected by its robustness and its recovery rapidity under this event, and in the case of sufficient repair resources, its resilience is mainly affected by the robustness. This paper simply regards the resilience of a CIS under an event as its robustness, which is quantified as system performance level immediately after the event and before any restoration efforts. Before presenting the mathematical formulation, some assumptions and simplifications are stated as follows:

- (1) The set of CISs of concern is denoted by κ . Each CIS k in κ is modeled by a network $G^k(N^k, E^k)$, described by a collection of nodes N^k and edges E^k . In CIS network k , there is a set of source nodes N^S , $k \subseteq N^k$ and a set of demand nodes N^D , $k \subseteq N^k$. Each node $n \in N^k$ is associated with s_n^k and d_n^k , representing its real commodity supply and its real satisfied demand, respectively. $\hat{s}_n^k$ is the maximum commodity supply and $\hat{d}_n^k$ is the required demand for node n (In the CIS operation, due to network capacity constraint, the real commodity supply of a node is bounded by its maximal output, while its real satisfied demand might be less than its required demand). If $\hat{s}_n^k > 0$ and $\hat{d}_n^k = 0$, node n in CIS network k is a source node; if $\hat{s}_n^k = 0$ and $\hat{d}_n^k = 0$, node n is a transshipment node; if $\hat{s}_n^k = 0$ and $\hat{d}_n^k > 0$, node n is a demand node; if $\hat{s}_n^k > 0$ and $\hat{d}_n^k > 0$, node n is both a source and a demand node. A variable f_e^k represents the flow on edge e in CIS network k , associated with its capacity $\hat{f}_e^k$. $o^k(e)$ and $d^k(e)$ denote the origin and destination nodes of edge e in CIS network k respectively.

For each CIS network and for illustrative purposes, this paper simply defines its performance level under a disruptive event (resilience metric in this paper) as the normalized total satisfied demand level $\sum_{n \in N^D, k} d_n^k / \sum_{n \in N^D, k} \hat{d}_n^k$. Then, the resilience of all CISs under this event is described by the weighted sum of the resilience of each CIS network, as expressed by [Eq. \(1\)](#), where η^k is the resilience weighting factor for CIS network k . Note that other performance metrics, such as those related to the operational cost can be also used, and the proposed approach is still valid.

- (2) Identifying the worst-case SLA and optimizing system resilience against the worst-case SLA need to introduce a virtual attacker and a defender. The attacker tries to disrupt the system by using the most harmful attack strategy (or the most beneficial attack strategy with respect to the attack expense), and the defender seeks the ex-ante actions (such as protect vulnerable components, build new edges to

increase redundancy) and ex-post actions (such as manipulate component flow) to maximize the resilience. The interactions between the attacker and the defender lead to a tri-level defender–attacker–defender model. In this model, the inner level manipulates the component flow to maximize the resilience, the middle level identifies the most harmful SLA, and the outer level makes the optimal resilience investment.

For the defender, there are many strategies that he/she can use to improve CIS resilience, as summarized in the literature ([Ouyang, 2014](#); [Ouyang, Dueñas-Osorio, & Min, 2012](#)). For illustrative purposes, this paper considers two typical ex-ante defense strategies, which have been also considered by other scholars to defend the intentional attacks in the literature ([Alderson et al., 2010](#); [Brown et al., 2006 & 2015](#)). First, assume the defender has the capacity to protect vulnerable components (nodes or edges or both) so that they are invulnerable to damage, which means that protected components within the attack area remain operating. Second, assume the defender has the capacity to build new edges and that any newly built edges will be invulnerable to attack. Note that the first strategy that protects weak components and the second strategy that can increase system redundancy, both can increase system capacity and then improve the total satisfied demand under disruptive events. Define binary variables $w_n^{P,k}$ and $w_e^{P,k}$ to represent whether the defender protects node n and edge e in CIS network k , and a binary variable $w_e^{B,k}$ to indicate whether the defender builds the candidate new line $e \in E^{A,k}$. The costs of protecting node n , protecting the line e and building the new line e in CIS network k are respectively denoted by $c_n^{P,k}$, $c_e^{P,k}$ and $c_e^{B,k}$. The defense budget is denoted by B_D , as described by [Constraint \(2\)](#). [Constraints \(3\)](#) and [\(4\)](#) enforce binary defense.

For the attacker, he/she will make spatially localized attacks on interdependent CISs. Modeling SLAs requires all CISs' geographical layout information. The locations of nodes can be described by xy-coordinates for systems mapped in a two-dimensional plate or latitude and longitude for systems described in the earth surface; the edges are assumed to be straight lines between two adjacent nodes. Note that if some edges in practice are not non-straight, we can divide them into short segments and each of them is a straight edge, then the proposed approach can still work. For the shape of the attack area, this paper mainly considers a circle shaped localized area, which is determined by a center Δ and a radius r_A . In addition, according to the SLAs-related events introduced in [Section 1](#), most of these events only caused one localized attack area, and as an initial step, this paper only considers one single attack. Also, in order to identify the worst-case attack, this paper assumes that in the presence of protection, the attacker can see which components have been protected or built before the attack. Among all components, a set of nodes $\tilde{N}^k \subseteq N^k$ and a set of lines $\tilde{E}^k \subseteq E^k$ in CIS network k can only be attacked by the attacker. Define $x_n^{N,k}$, $x_e^{E,k}$ as the normal/damage state of a CIS node and a CIS edge. If the shortest distance between a node or an edge and the attack center Δ is less than or equal to r_A , then this component will fail, i.e., $x_n^{N,k} = 0$ or $x_e^{E,k} = 0$; if the component is protected, the component will be invulnerable, i.e., $x_n^{N,k} = 1$ or $x_e^{E,k} = 1$. This SLAs modeling is described by [Eqs. \(5\)–\(8\)](#). [Constraints \(5\)–\(6\)](#) determine the component damage states, and only components with their shortest distances $dist$ from the attack center Δ not larger than r_A would fail. When the component is a node, the shortest distance between the attack center and this node can

be directly computed based on their coordinates. When the component is an edge, the shortest distance between the attack center and this edge can be computed by checking the position of the foot of perpendicular from the attack center to the line including the edge. If the position of the foot of perpendicular is inside the edge, the shortest distance is the distance between the attack center and the foot of perpendicular; if the position of the foot of perpendicular is outside the edge, the shortest distance is the minimum distance between the attack center and the two endpoints of the edge. Constraint (7) forces the damage state of each invulnerable component as zero. Constraint (8) ensures that each protected node or edge cannot be destroyed.

- (3) The operation of each CIS network is modeled by a network flow model, which has been frequently used to model general CISs (Gonzalez, Duenas-Orsorio, Sanchez-Silva, & Medaglia, 2016). In this model, the decision variables in CIS network k include the real commodity supply of each node s_n^k , the real satisfied demand of each node d_n^k and the flow in each line f_e^k . The constraints of these decision variables are described by Constraints (9)–(12). Constraint (9) guarantees flow conservation at each node. Constraint (10) states the maximum edge flows in each edge. Constraint (11) defines the minimum and the maximum commodity supply for each source node. Constraint (12) states that the real satisfied demand cannot exceed the required demand for each load node.

CISs might be physically, cyber, geographically, logically interdependent, where physical interdependency means the state of one CIS depends on the material output(s) of the other; cyber interdependency means the state of one CIS depends on information transmitted by the information infrastructure; geographical interdependency means a local environmental event can create state changes in multiple CISs; logical interdependency means the state of each CIS depends on the state of the other via a mechanism that is not a physical, cyber, or geographic connection (Rinaldi et al., 2001). As the cyber interdependency needs to additionally consider the information infrastructure, for simplification, this paper will mainly consider other three types of interdependencies. Note that if integrating the information infrastructure into the framework, cyber interdependency can be modeled similarly to the physical interdependency and the proposed approach is still valid.

The physical interdependency can be modeled by defining a set of ordered component pairs (i, j) associated with node i in one CIS network and component (node or edge) j in another CIS network, where component j is operational if node i can provide a fully supplied demand of commodities (Coffrin, Hentenryck, & Bent, 2012; Gonzalez et al., 2016; Nurre, Cavdaroglu, Mitchell, & Sharkey, 2012). Define $C^k(G^k)$ as the set of all nodes (all edges) in CIS network k which depend on other CIS nodes to operate. Define $C^{k \leftarrow l}(G^{k \leftarrow l})$ as the set of all nodes (all edges) in CIS network k that depend on the nodes in CIS network l to operate. Define D^k as the set of all nodes in CIS network k that any other CIS networks depend on. Define $D^{k \rightarrow l}$ as the set of all nodes in CIS network k that the CIS network l depends on. Define $F_{i,j}^{k \rightarrow l}$ as the set of order pairs associated with node i in $D^{k \rightarrow l}$ and node j in $C^{(l \leftarrow k)}$, and $M_{i,j}^{k \rightarrow l}$ as the set of order pairs associated with node i in $D^{k \rightarrow l}$ and edge j in $G^{l \leftarrow k}$. Define $\delta_{i,j}^{k \rightarrow l}$ as a binary variable, which is 1 if the physical interdependency from node i in CIS network k to component j in CIS network l can work normally, and 0 otherwise. Based on the above defined input parameters and decision

variables, the physical interdependencies between multiple CISs can be described by Constraints (13)–(17). For each ordered pair $(i, j) \in F_{i,j}^{k \rightarrow l} \cup M_{i,j}^{k \rightarrow l}$, the demand level for node i in CIS network k should be either zero or full level, depending on whether their interdependency relationship can work normally ($\delta_{i,j}^{k \rightarrow l} = 1$ for normal operation, and 0 otherwise), as stated by Constraint (13). Also, for each component j in the ordered pair $(i, j) \in F_{i,j}^{k \rightarrow l} \cup M_{i,j}^{k \rightarrow l}$, depending on the type of this component, it has different constraints. If component j is a source node, its output level is bounded by zero or the maximal output, as stated by Constraint (14). If component j is a demand node, its demand level is bounded by zero or the required demand, as stated by Constraint (15). If component j is a node and it is damaged, all its attached edges will fail and the flow on these edges should be also zero, as stated by Constraint (16). If component j is an edge, its flow depends on its normal/damage state, as stated by Constraint (17).

For the geographical interdependency, this paper considers multiple CISs located within the same region, and the SLAs cause the simultaneous failure of all unprotected CISs' components distributed over the attack area, as described by Constraints (5)–(6). The modeling of the logical interdependency depends on the specific system of concern and the system environment. This paper considers a simple type of logical interdependency for illustrative purposes. For each order pair (i, j) with physical interdependency, if interdependent demand component j is damaged, interdependent source node i should have its real demand as zero. In this case, the interdependency relationship $(i, j) \in F_{i,j}^{k \rightarrow l} \cup M_{i,j}^{k \rightarrow l}$ does not normally work and the physical interdependency variable $\delta_{i,j}^{k \rightarrow l}$ becomes zero, as described by Constraints (18)–(19).

Based on the above description, the model for resilience optimization of interdependent CISs against spatially localized attacks (ROICISs-SLAs) is formulated as follows:

Input Parameters for each CIS Network

$k \in \mathcal{K}$	set of all CIS networks
$n \in \mathcal{N}^k$	set of nodes in CIS network k
$e \in \mathcal{E}^k$	set of edges in CIS network k
$e \in \mathcal{E}^{A,k}$	set of additional edges available to be built in CIS network k , $\mathcal{E}^k \cap \mathcal{E}^{A,k} = \emptyset$
$n \in \mathcal{N}^{S,k}$	set of source nodes in CIS network k
$n \in \mathcal{N}^{D,k}$	set of demand nodes in CIS network k
$e \in \mathcal{E}_n^{I,k}$	set of edges in $\mathcal{E}^k \cup \mathcal{E}^{A,k}$ oriented to node n in CIS network k
$e \in \mathcal{E}_n^{O,k}$	set of edges in $\mathcal{E}^k \cup \mathcal{E}^{A,k}$ oriented from node n in CIS network k
$\tilde{\mathcal{N}}^k \subseteq \mathcal{N}^k$	set of nodes that can be damaged in CIS network k
$\tilde{\mathcal{E}}^k \subseteq \mathcal{E}^k$	set of edges that can be damaged in CIS network k
$o^k(e), d^k(e)$	origin and destination nodes of edge e in CIS network k
$\hat{f}_e^k$	capacity for edge e in CIS network k
$\hat{s}_n^k$	maximum output for node n in CIS network k
$\hat{d}_n^k$	required demand for node n in CIS network k

Input Parameters for CISs Interdependencies

C^k	set of all nodes in CIS network k which depend on other CIS nodes to operate
G^k	set of all edges in CIS network k which depend on other CIS nodes to operate
D^k	set of all nodes in CIS network k which any other CIS nodes depend on
$C^{k \leftarrow l}$	set of all nodes in CIS network k which depend on CIS network l to operate

$G^{k \leftarrow l}$ set of all edges in CIS network k which depend on CIS network l to operate
 $D^{k \rightarrow l}$ set of all nodes in CIS network k which CIS network l depends on
 $F_{i,j}^{k \rightarrow l}$ set of ordered pairs (i, j) associated with node i in $D^{k \rightarrow l}$ and node j in $G^{k \leftarrow l}$, and node j is operational when node i can provide a fully supplied demand of commodities in CIS network k
 $M_{i,j}^{k \rightarrow l}$ set of ordered pairs (i, j) associated with node i in $D^{k \rightarrow l}$ and edge j in $G^{k \leftarrow l}$, and component j is operational when node i can provide a fully supplied demand of commodities in CIS network k

Input Parameters for attack capacity of the attacker
 r_A circle radius of the spatially localized attack

Input Parameters for defense capacity and cost of the defender
 $c_n^{P,k}$ cost to protect node n in CIS network k
 $c_e^{P,k}$ cost to protect edge e in CIS network k
 $c_e^{B,k}$ cost to build additional edge e in CIS network k
 B_D budget for resilience enhancement

Input Parameters for resilience weighting factor of each CIS network
 η^k resilience weighting factor for CIS network k

Ex-ante decision variables for the defender
 $w_n^{P,k}$ 1 if defender protects node n in CIS network k , 0 otherwise
 $w_e^{P,k}$ 1 if defender protects edge e in CIS network k , 0 otherwise
 $w_e^{B,k}$ 1 if defender builds additional e in CIS network k , 0 otherwise

Decision variables for the attacker
 Δ location of attack center
 $x_n^{N,k}$ damage state of node n in CIS network k under a given damage scenario, taking the value 0 if node is damaged and 1 otherwise.
 $x_e^{E,k}$ damage state of edge e in CIS network k under a given damage scenario, taking the value 0 if the line is damaged and 1 otherwise.

Ex-post decision variables for the defender
 s_n^k output from node n in CIS network k
 d_n^k demand for demand node n in CIS network k
 f_e^k flow on edge e in CIS network k
 $\delta_{i,j}^{k \rightarrow l}$ 1 if the interdependency from node i in CIS network k to component j in CIS network l works normally

Formulation of ROICISs-SLAs Model:

$$\max_{w_n^{P,k}, w_e^{P,k}, w_e^{B,k}, x_n^{N,k}, x_e^{E,k}, \Delta} \min_{s_n^k, d_n^k, f_e^k, \delta_{i,j}^{k \rightarrow l}} R = \sum_{k \in \mathcal{K}} \eta^k \sum_{n \in \mathbf{N}^{D,k}} d_n^k / \sum_{n \in \mathbf{N}^{D,k}} \hat{d}_n^k \quad (1)$$

Subject to:

First stage (design) constraints:

$$\sum_{k \in \mathcal{K}} \sum_{n \in \tilde{N}^k} c_n^{P,k} w_n^{P,k} + \sum_{k \in \mathcal{K}} \sum_{e \in E^{A,k}} c_e^{B,k} w_e^{B,k} + \sum_{k \in \mathcal{K}} \sum_{e \in E^{A,k}} c_e^{P,k} w_e^{P,k} \leq B_D \quad (2)$$

$$w_e^{B,k} \in \{0, 1\}, \forall e \in E^{A,k}, \forall k \in \mathcal{K} \quad (3)$$

$$w_n^{P,k}, w_e^{P,k} \in \{0, 1\}, \forall e \in \tilde{E}^k, \forall n \in \tilde{N}^k, \forall k \in \mathcal{K} \quad (4)$$

Second stage (attack) constraints:

$$x_n^{N,k} = \begin{cases} w_n^{P,k} & \text{dist}(n, \Delta) \leq r_A \\ 1 & \text{dist}(n, \Delta) > r_A \end{cases}, \quad n \in \tilde{N}^k \quad (5)$$

$$x_e^{E,k} = \begin{cases} w_e^{P,k} & \text{dist}(e, \Delta) \leq r_A \\ 1 & \text{dist}(e, \Delta) > r_A \end{cases} \quad \forall e \in \tilde{E}^k \quad (6)$$

$$x_n^{N,k} = 1, x_e^{E,k} = 1, \forall k \in \mathcal{K}, \forall n \in N^k \setminus \tilde{N}^k, \forall e \in E^k \setminus \tilde{E}^k \quad (7)$$

$$x_e^{E,k} = w_e^{B,k}, \quad \forall e \in E^{A,k} \quad (8)$$

Third stage (response/recovery, multi-period) constraints:

$$s_n^k + \sum_{(e \in E^k \cup E^{A,k} | d^k(e)=n)} f_e^k - \sum_{(e \in E^k \cup E^{A,k} | o^k(e)=n)} f_e^k = d_n^k, \quad \forall n \in N^k \quad (9)$$

$$-f_e^k x_e^{E,k} x_{o^k(e)}^{N,k} x_{d^k(e)}^{N,k} \leq f_e^k \leq x_e^{E,k} x_{o^k(e)}^{N,k} x_{d^k(e)}^{N,k} \hat{f}_e^k, \quad \forall e \in E^k \cup E^{A,k} \quad (10)$$

$$0 \leq s_n^k \leq x_n^{N,k} \hat{s}_n^k, \quad \forall n \in N^{S,k} \quad (11)$$

$$0 \leq d_n^k \leq x_n^{N,k} \hat{d}_n^k, \quad \forall n \in N^{D,k} \quad (12)$$

$$d_i^k = \delta_{i,j}^{k \rightarrow l} \hat{d}_i^k, \quad \forall (i, j) \in F_{i,j}^{k \rightarrow l} \cup M_{i,j}^{k \rightarrow l} \quad (13)$$

$$s_j^l \leq \delta_{i,j}^{k \rightarrow l} \hat{s}_j^l, \quad \forall (i, j) \in F_{i,j}^{k \rightarrow l} \quad (14)$$

$$d_j^k \leq \delta_{i,j}^{k \rightarrow l} \hat{d}_j^k, \quad \forall (i, j) \in F_{i,j}^{k \rightarrow l} \quad (15)$$

$$f_e^k \leq \delta_{i,j}^{k \rightarrow l} \hat{f}_e^k, \quad \forall (i, j) \in F_{i,j}^{k \rightarrow l}, \forall e \in E_{i,j}^{l,k} \cup E_j^{O,k} \quad (16)$$

$$f_j^k \leq \delta_{i,j}^{k \rightarrow l} \hat{f}_j^k, \quad \forall (i, j) \in M_{i,j}^{k \rightarrow l} \quad (17)$$

$$\delta_{i,j}^{k \rightarrow l} \leq x_j^{N,l}, \quad \forall (i, j) \in F_{i,j}^{k \rightarrow l} \quad (18)$$

$$\delta_{i,j}^{k \rightarrow l} \leq x_j^{E,l}, \quad \forall (i, j) \in M_{i,j}^{k \rightarrow l} \quad (19)$$

Note that the ROICISs-SLAs model is a non-linear mixed integer programming (MIP) problem, which can be exactly solved by a decomposition algorithm introduced in the next section.

3. Solution algorithm

If denote all CISs' operational decision variables $s_n^k, d_n^k, \delta_{i,j}^{k \rightarrow l}, f_e^k$ by a single vector $\mathbf{y}$, the attack decision variables $x_e^{E,k}, x_n^{N,k}, \Delta$ by a single vector $\mathbf{x}$, and the defense decision variables $w_n^{P,k}, w_e^{P,k}, w_e^{B,k}$ by a single vector $\mathbf{w}$, then the proposed ROICISs-SLAs model can be simply described by $\max_{\mathbf{w}} \min_{\mathbf{x}} \max_{\mathbf{y}} R(\mathbf{w}, \mathbf{x}, \mathbf{y})$. This section presents a decomposition algorithm to solve this model. The proposed algorithm is based on the decomposition algorithm proposed by Alderson et al. (2010 & 2014), which is useful for non-proximity based intentional attacks that any components can be attacked. This paper adapts this algorithm for the spatially localized attacks. The proposed algorithm begins by solving a sub-problem $\min_{\mathbf{x}} \max_{\mathbf{y}} R(\hat{\mathbf{w}}, \mathbf{x}, \mathbf{y})$, which solves the worst-case attack under a given defense strategy $\hat{\mathbf{w}}$. If $\hat{\mathbf{w}}$ is just the optimal solution $\mathbf{w}^*$ for the original problem $\max_{\mathbf{w}} \min_{\mathbf{x}} \max_{\mathbf{y}} R(\mathbf{w}, \mathbf{x}, \mathbf{y})$, then solving the sub-problem will return the optimal objective value R_{Sp}^* equal to the optimal objective value R^* for the original problem. However, if $\hat{\mathbf{w}}$ is not $\mathbf{w}^*$, then solving the sub-problem will return the optimal objective value R_{Sp}^* less than R^* . Hence, for any fixed defense strategy $\hat{\mathbf{w}}$, solving the sub-problem returns a lower bound of R^* . Next, the algorithm solves a master problem $\max_{\mathbf{w}} \max_{\{\mathbf{y}^h\}} R(\mathbf{w}, \{\hat{\mathbf{x}}^h\}, \{\mathbf{y}^h\})$, which solves the optimal defense strategy given a fixed set of attack strategies $\{\hat{\mathbf{x}}^h\}_{h=1,2,\dots,t}$ identified from the

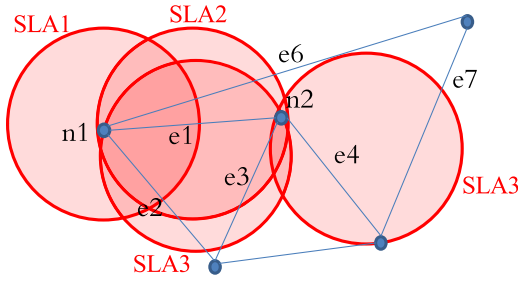


Fig. 1. A simple network and four SLA scenarios with the same attack radius. The SLA1 covers components $n1, e1, e2, e6$; the SLA2 covers a maximal component set $\{n1, n2, e1, e2, e3, e4, e6\}$; the SLA3 covers the maximal component set $\{n1, n2, e1, e2, e3, e4, e6\}$, with the circle boundary passing through two nodes $n1$ and $n2$; the SLA4 covers another maximal component set $\{n2, n4, e1, e3, e4, e5, e6, e7\}$, with its circle boundary tangent to an edge $e6$ and passing through node $n4$.

sub-problem. If the set of attack strategies $\{\hat{x}^h\}$ covers all possible attack strategies, then solving the master problem will return the optimal objective value R_{MP}^* equal to R^* ; if the set of attack strategies $\{\hat{x}^h\}$ is a partial of all possible attack strategies, solving the master problem will return the optimal objective value R_{MP}^* larger than R^* . Hence, for any fixed set of attack strategies $\{\hat{x}^h\}$, solving the master problem returns an upper bound of R^* . By solving the sub-problem and the master problem iteratively, the set of attack strategies $\{\hat{x}^h\}$ enlarges and the gap between the upper and the lower bound of R^* will decrease into a small acceptable value so that the original problem is efficiently solved. This section next introduces the solution algorithms for the above two problems.

3.1. Solution algorithm for the sub-problem $\min_x \max_y R(\hat{w}, x, y)$

Assume that at the iteration t of the proposed algorithm, a specific defense strategy $\hat{w} = \hat{w}^t$ is given, solving the sub-problem yields the worst-case attack. This problem needs to identify a set of components covered by a circle with the radius r_A , and cannot be solved by any existing algorithm. This section next introduces how to exactly solve this sub-problem based on the concept of the “maximal component set” (Ouyang, 2016; Ouyang et al., 2017).

This paper next takes a simple network and four circle shaped attack scenarios with the same r_A , as displayed in Fig. 1, to illustrate the solution concept for the sub-problem. As the attack center can be in any place on the system map, the number of possible attack scenarios is numerous. The SLA1 in the figure is a typical attack scenario, which covers the components $n1, e1, e2, e6$, but note that this attack scenario can be moved into some position like the SLA2 to cover three more components $n2, e3, e4$. Because it can be proved that the failure of a set of system components will cause the resilience not larger than the failure of any subset of these components¹, it means that the SLA2 is a worse attack scenario than the SLA1. The next question is whether the SLA2 can be further moved to some position to cover more components. The answer is negative, then these seven components $\{n1, n2, e1, e2, e3, e4, e6\}$ covered by the circle in the SLA2 can be called a maximal component set (MCS), which is defined as the set of all compo-

nents that can be covered by a circle with radius r_A and adding any new component will make them not able to be covered. Based on this definition, the worst-case SLA must cover a MCS. Hence, solving the sub-problem can be converted into identifying all MCSs.

Paying attention to the attack scenario SLA2 again, note that this attack scenario can be moved into some position like the SLA3 with its circle boundary passing through nodes $n1$ and $n2$. The SLA4 is another attack scenario, also covering a maximal component set $\{n2, n4, e1, e3, e4, e5, e6, e7\}$ and with its circle boundary tangent to edge $e6$ and passing through node $n4$. From these two attack scenarios SLA3 and SLA4, it can be found that any MCS can be covered by a circle with its circle boundary passing through or tangent to two components. Based on this finding, all MCSs can be identified according to the following procedures: (1) Given the attack radius r_A , for any two system components (two nodes, or one node and one edge, or two edges), find out the circle center with its circle boundary passing through or tangent to these two components. For a network with $|N| + |E|$ components (nodes and edges), there are $(|N| + |E|)(|N| + |E| - 1)/2$ pairs of components, but searching the attack centers does not need to consider all these component pairs. If the distance between two components is larger than $2r_A$, these two components cannot be covered by a circle with radius r_A , then this component pair is not considered for further analysis as the attack center does not exist; if their distance is not larger than $2r_A$, the attack center can be found according to Equations in Ouyang, 2016. (2) For each attack center, compute the shortest distance between the attack center and each component (node or edge) to identify whether the component can be covered by a circle with radius r_A , and then these components make up a candidate MCS. (3) Refine all candidate MCSs so that each component set is not the proper subset of any other component set, then the refined candidate MCSs is just all MCSs.

After identifying all MCSs, for each of them, the failure of its contained components will produce a fixed value $\hat{x}$, and then the sub-problem becomes the inner-most operator model $\max_y R(\hat{w}, \hat{x}, y)$, subjected to Constraints (9)–(19). Note that given $\hat{w}, \hat{x}$, all constraints (9)–(19) are linear, while s_n^k, d_n^k, f_e^k are continuous decision variables, while $\delta_{i,j}^{k \rightarrow l}$ is binary decision variable, so the inner-most operator model is a linear MIP model and can be directly solved by CPLEX. The optimum MCS that causes the least resilience, which is just the optimal objective value of the sub-problem R_{sp}^* , can be found. The worst-case attack scenario will cover this optimum MCS, and the attack center Δ can be determined by a series of inequalities, which describe the shortest distance between Δ and each component in the optimum MCS is less than or equal to r_A .

3.2. Solution algorithm for the master-problem $R(w, \{\hat{x}^h\}, \{y^h\})$

Assume that at the iteration t of the proposed algorithm, a set of attack scenarios $\{\hat{x}^h\}_{h=1,2,\dots,t}$ is obtained from solving the previous iterations of the sub-problem. The master problem maximizes the resilience under these attack scenarios and its solution yields an optimal defense strategy $\hat{w}^t$ and a set of optimal operational strategies $\hat{y}^h, h = 1, 2, \dots, t$ to respond to each of these attacks $\hat{x}^h$. The master problem can be formulated as:

$$\max_{z, w, \{y^h\}} z \quad (20)$$

Subject to:

$$z \leq \sum_{k \in K} \sum_{n \in N^{D,k}} \eta^k d_n^k(h), \forall h = 1, 2, \dots, t \quad (21)$$

(2)–(4)

$$s_n^k(h) + \sum_{(e \in E^k \cup E^{A,k} | d^k(e)=n)} f_e^k(h) - \sum_{(e \in E^k \cup E^{A,k} | o^k(e)=n)} f_e^k(h)$$

¹ Denote the set of all system components by I . Given the failure of a set A of system components, the optimal flow on system components to maximize the resilience can be found by solving the inner-level ROICIS-SLAs model, note that these flow $f(A)$ on system components $I-A$ all satisfy the operational constraints (9)–(19). For the failure of any subset B of A , a possible solution for component flow can be obtained by assigning the flow data $f(A)$ to system components $I-A$ and assigning zero flow to system components $A-B$, and then the resilience under the failure of B must be not less than that under the failure of A , which indicates the failure of a set of components will cause the resilience not larger than the failure of a subset of these components.

$$= d_n^k(h), \forall n \in N^k, \forall h = 1, 2, \dots, t \quad (22)$$

$$\begin{aligned} & -\hat{f}_e^k [\hat{x}_e^{E,k}(h) + (1 - \hat{x}_e^{E,k}(h))w_e^{P,k}] [\hat{x}_{o^k(e)}^{N,k}(h) + (1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k}] \\ & \times [\hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k}] \\ & \leq f_e^k(h) \leq \hat{f}_e^k [\hat{x}_e^{E,k}(h) + (1 - \hat{x}_e^{E,k}(h))w_e^{P,k}] \\ & \times [\hat{x}_{o^k(e)}^{N,k}(h) + (1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k}] \\ & \times [\hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k}], \\ & \forall e \in E^k, \forall h = 1, 2, \dots, t \end{aligned} \quad (23)$$

$$\begin{aligned} & -\hat{f}_e^k w_e^{B,k} [\hat{x}_{o^k(e)}^{N,k}(h) + (1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k}] \\ & \times [\hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k}] \leq f_e^k(h) \leq \hat{f}_e^k w_e^{B,k} \\ & \times [\hat{x}_{o^k(e)}^{N,k}(h) + (1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k}] \\ & \times [\hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k}], \\ & \forall e \in E^{A,k}, \forall h = 1, 2, \dots, t \end{aligned} \quad (24)$$

$$\begin{aligned} 0 & \leq s_n^k(h) \leq [\hat{x}_n^{N,k}(h) + (1 - \hat{x}_n^{N,k}(h))w_n^{P,k}] \hat{s}_n^k, \\ & \forall n \in N^{S,k}, \forall h = 1, 2, \dots, t \end{aligned} \quad (25)$$

$$\begin{aligned} 0 & \leq d_n^k(h) \leq [\hat{x}_n^{N,k}(h) + (1 - \hat{x}_n^{N,k}(h))w_n^{P,k}] \hat{d}_n^k, \\ & \forall n \in N^{D,k}, \forall h = 1, 2, \dots, t \end{aligned} \quad (26)$$

$$d_i^k(h) = \delta_{i,j}^{k \rightarrow l}(h) \hat{d}_i^l, \forall (i, j) \in F_{i,j}^{k \rightarrow l} \cup M_{i,j}^{k \rightarrow l}, \forall h = 1, 2, \dots, t \quad (27)$$

$$s_j^l(h) \leq \delta_{i,j}^{k \rightarrow l}(h) \hat{s}_j^l, \forall (i, j) \in F_{i,j}^{k \rightarrow l}, \forall h = 1, 2, \dots, t \quad (28)$$

$$d_j^k(h) \leq \delta_{i,j}^{k \rightarrow l}(h) \hat{d}_j^l, \forall (i, j) \in F_{i,j}^{k \rightarrow l}, \forall h = 1, 2, \dots, t \quad (29)$$

$$f_e^k(h) \leq \delta_{i,j}^{k \rightarrow l}(h) \hat{f}_e^k, \forall (i, j) \in F_{i,j}^{k \rightarrow l}, \forall e \in E_{i,j}^{l,k} \cup E_j^{O,k}, \forall h = 1, 2, \dots, t \quad (30)$$

$$f_j^k(h) \leq \delta_{i,j}^{k \rightarrow l}(h) \hat{f}_j^k, \forall (i, j) \in M_{i,j}^{k \rightarrow l}, \forall h = 1, 2, \dots, t \quad (31)$$

$$\delta_{i,j}^{k \rightarrow l}(h) \leq \hat{x}_j^{N,l}(h) + (1 - \hat{x}_j^{N,l}(h))w_j^{P,l}, \forall (i, j) \in F_{i,j}^{k \rightarrow l}, \forall h = 1, 2, \dots, t \quad (32)$$

$$\delta_{i,j}^{k \rightarrow l}(h) \leq \hat{x}_j^{E,l}(h) + (1 - \hat{x}_j^{E,l}(h))w_j^{P,l}, \forall (i, j) \in M_{i,j}^{k \rightarrow l}, \forall h = 1, 2, \dots, t \quad (33)$$

This master problem has linear objective function and the constraints are almost all linear except some non-linear binary product terms, such as $[\hat{x}_e^{E,k}(h) + (1 - \hat{x}_e^{E,k}(h))w_e^{P,k}][\hat{x}_{o^k(e)}^{N,k}(h) + (1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k}][\hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k}]$. These non-linear terms can be linearized according to McCormick method. By using this technique, introducing a new variable $\eta_e^k(h) = [\hat{x}_e^{E,k}(h) + (1 - \hat{x}_e^{E,k}(h))w_e^{P,k}][\hat{x}_{o^k(e)}^{N,k}(h) +$

$(1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k}] [\hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k}]$ for each edge makes Constraint (23) equivalent to the following linear constraints:

$$-\hat{f}_e^k \eta_e^k(h) \leq f_e^k(h) \leq \eta_e^k(h) \hat{f}_e^k \quad (34)$$

$$0 \leq \eta_e^k(h) \leq \hat{x}_e^{E,k}(h) + (1 - \hat{x}_e^{E,k}(h))w_e^{P,k} \quad (35)$$

$$0 \leq \eta_e^k(h) \leq \hat{x}_{o^k(e)}^{N,k}(h) + (1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k} \quad (36)$$

$$0 \leq \eta_e^k(h) \leq \hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k} \quad (37)$$

$$\begin{aligned} & \hat{x}_e^{E,k}(h) + (1 - \hat{x}_e^{E,k}(h))w_e^{P,k} + \hat{x}_{o^k(e)}^{N,k}(h) + (1 - \hat{x}_{o^k(e)}^{N,k}(h))w_{o^k(e)}^{N,k} \\ & + [\hat{x}_{d^k(e)}^{N,k}(h) + (1 - \hat{x}_{d^k(e)}^{N,k}(h))w_{d^k(e)}^{N,k}] - 2 \leq \eta_e^k(h) \leq 1 \end{aligned} \quad (38)$$

Other binary product terms can also be converted into linear constraints in a similar manner, and then the master problem will become a linear MIP model, which can also be solved by CPLEX.

According to the algorithms for the above sub-problem and the master-problem, a description of the algorithm to solve the original ROICIS-SLAs model follows:

Input: Full ROICIS-SLAs problem data, the iteration limit $t^{\max} > 0$ and the optimality tolerance ϵ for the decomposition.

Output: the ϵ -optimal iteration-limited defense strategy $\mathbf{w}^*$ and the worst attack scenario $\mathbf{x}^*$.

- (1) Identify all maximal component sets MCSs given r_A ; set $t \leftarrow 1$, $\hat{\mathbf{w}}^0 \leftarrow \emptyset$, $R^{\text{UP}} \leftarrow \infty$, $R^{\text{LO}} \leftarrow 0$.
- (2) Solve the sub-problem: set $R_m^{\text{t}, \min} \leftarrow 0$, $\hat{\mathbf{x}}^t \leftarrow \emptyset$; for each MCS_m, determine the component set that can be damaged based on Constraints (5)–(8), and compute its failure $\hat{\mathbf{x}}_m^t$ induced resilience R_m^t according to the inner-most model $R(\mathbf{w}^t, \hat{\mathbf{x}}_m, \mathbf{y})$, if $R_m^t < R_m^{\text{t}, \min}$, set $R_m^{\text{t}, \min} \leftarrow R_m^t$, $\hat{\mathbf{x}}^t \leftarrow \hat{\mathbf{x}}_m^t$.
- (3) If $R_m^{\text{t}, \min} > R^{\text{LO}}$, set $R^{\text{LO}} \leftarrow R_m^{\text{t}, \min}$.
- (4) Solve the master-problem: given the attack scenarios $\hat{\mathbf{x}}^1, \hat{\mathbf{x}}^2, \dots, \hat{\mathbf{x}}^t$, solve the master problem to determine the optimal defense strategy $\hat{\mathbf{w}}^t$ and the optimal objective value z^* .
- (5) If $z^* < R^{\text{UP}}$, set $R^{\text{UP}} \leftarrow z^*$.
- (6) If $R^{\text{UP}} - R^{\text{LO}} \leq \epsilon$ or $t > t^{\max}$, go to the step (8).
- (7) $t = t + 1$, go to the step (2).
- (8) End: the best defense strategy is $\mathbf{w}^* = \hat{\mathbf{w}}^t$, and the worst attack $\mathbf{x}^*$ is obtained by running the step (2) with $\mathbf{w}^t = \mathbf{w}^*$.

4. Case study

This section presents numerical experiments on interdependent power and water systems. The power system is the IEEE 14 bus test system, and the water system is adapted according to the IEEE 9 bus system. Studying the spatially localized attacks needs the geographical layout information of each system. The author has assigned position information to each system node, as provided in Table 1. A geographical layout of these two systems is shown in Fig. 2. The edges are straight lines between adjacent nodes. All source nodes are assumed invulnerable and all other components can be damaged. Five potential edges available for construction are displayed by dashed lines in the figure. For the physical interdependency, the water node w8 depends on the power demand node p11; the water node w7 depends on the power demand node p10; the water node w1 depends on the power demand node p4; the water node w3 depends on the power demand node p9; the power generator node p1 depends on the water demand node w9.

Table 1
Coordinates of each node in the power and water systems.

Node ID	Coordinate X (kilometers)	Coordinate Y (kilometers)	Node ID	Coordinate X (kilometers)	Coordinate Y (kilometers)
p1	1	4	p13	5	8
p2	1	2	p14	9	8
p3	5	0	w1	5	2
p4	5	2	w2	2	7
p5	5	4	w3	9	4
p6	1	6	w4	3	2
p7	9	2	w5	7	0
p8	9	0	w6	7	4
p9	9	4	w7	7	6
p10	7	6	w8	4	6
p11	4	6	w9	1	4
p12	1	8	–	–	–

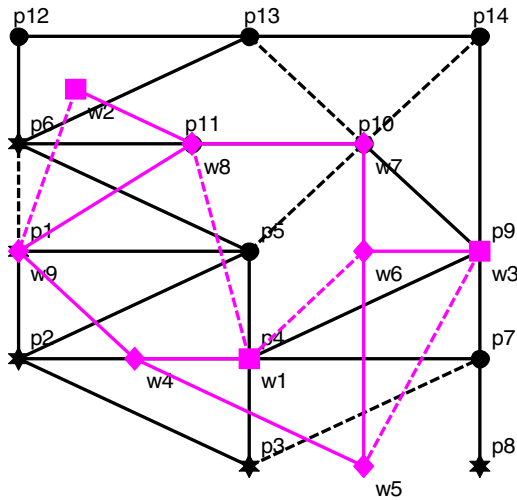


Fig. 2. A geographical layout of the interdependent power and water systems.

The experiments are performed on a laptop with Intel (R) Core (TM) 2 Due 2.5 Gigahertz CPU and 4 Gigabytes memory. The proposed algorithm is implemented on MATLAB with CPLEX MATLAB toolbox. The algorithm sets the number of iterations to be the number of all MCSs under a given r_A , and the acceptable gap of the optimum resilience is set as 0.001. In addition, this paper assumes that protecting one node in different CISs all needs two units of resources, protecting one edge in different CISs all needs one unit of resources, and building one new edge in different CISs needs two units of resources. The resilience weighting factor is set as 0.5 for each system.

4.1. Identification of the worst-case attack and the best defense strategy

First, when there is no defense, i.e., $\hat{w} = 0$, solving the proposed ROICISs-SLAs model produces the worst SLA under a given attack radius r_A . Fig. 3 shows the results for each single system and the interdependent systems under $r_A=0.5, 1.0$ and 2.5 kilometers. From the figure, the interdependencies influence the location of the worst-case SLA. For example, when $r_A=1.0$ kilometers, for single power system, the worst SLA causes the failure of the power nodes p4, p5 and the power lines p1–p5, p2–p4, p2–p5, p3–p4, p4–p5, p4–p7, p4–p9, p5–p6; for single water system, the worst-case SLA causes the failure of the water nodes w6, w7 and the water lines w3–w6, w5–w6, w6–w7, w7–w8; for the interdependent systems, the worst SLA causes the failure of the

power node p4, the power lines p2–p4, p3–p4, p4–p5, p4–p7, p4–p9 and the water lines w1–w4, w5–w6. The power components covered by the worst-case SLA for the interdependent systems is a subset of those for single power system, but there is only one water component (water line w5–w6) covered by both the worst-case SLA for the interdependent systems and the worst-case SLA for the water system. Besides the worst-case SLA, Fig. 3(d) also displays the resilience associated with the second-worst through the fifth-worst combination of resilience for the interdependent systems under different r_A . From the figure, it can be found that the worst-case SLA is unique because for each r_A , the resilience under the second-worst attack is always strictly larger than that under the worst-case SLA. Also, under some r_A , such as 0.5 kilometers, 2.5 kilometers, or 3.0 kilometers, the second-worst attack also causes small resilience with the value a little larger than that for the worst-case attack. This indicates that protecting interdependent systems only against the worst-case SLA might be insufficient.

Second, when the defense is considered, given the attack radius and a defense budget, solving the ROICISs-SLAs model can produce the best defense strategy and the worst-case SLA. Fig. 4 shows the results for each single system and the interdependent systems when $r_A=1.0$ kilometers and D_B is respectively set as 2, 5, 11 units. From the figure, the interdependencies affect the optimum defense strategy. For example, when $D_B=5$, for single power system, the optimum defense is to protect the power nodes p4, p9 and the power line p4–p9; for single water system, the optimum defense is to build the new water line w3–w5, and protect the water node w9 and the water line w4–w9; for the interdependent systems, the optimum defense is to build the new water lines w3–w5, w2–w9, and protect the water line w8–w9. The optimal defense strategies in the case of considering interdependencies are totally different from those for each single system. Also, increasing the defense budget affects the optimum defense strategy and the worst-case SLA, as also shown in the figure. In addition, given the optimum defense $(\hat{w}_n^{P,k}, \hat{w}_e^{P,k}, \hat{w}_e^{B,k})$ identified from the proposed algorithm, its uniqueness can be checked by adding the following constraint in the master problem to eliminate the existing solution: $\sum_k \sum_{n: \hat{w}_n^{P,k}=0} w_n^{P,k} + \sum_k \sum_{n: \hat{w}_n^{P,k}=1} (1 - w_n^{P,k}) + \sum_k \sum_{e: \hat{w}_e^{P,k}=0} w_e^{P,k} + \sum_k \sum_{e: \hat{w}_e^{P,k}=1} (1 - w_e^{P,k}) + \sum_k \sum_{e: \hat{w}_e^{B,k}=0} w_e^{B,k} + \sum_k \sum_{e: \hat{w}_e^{B,k}=1} (1 - w_e^{B,k}) \geq 1$. When $r_A=1.0$ and $D_B=5$, besides the optimum defense strategy shown in Fig. 4(h), there also exist several other equivalent optimum solutions, including (1) build the water lines w3–w5, w2–w9, protect the water line w7–w8; (2) build the water lines w3–w5, w2–w9, protect the water line w5–w6; (3) build the water lines w3–w5, w2–w9, protect the power line p6–p13; (4) build the water lines w3–w5, w2–w9, protect the power line p6–p11.

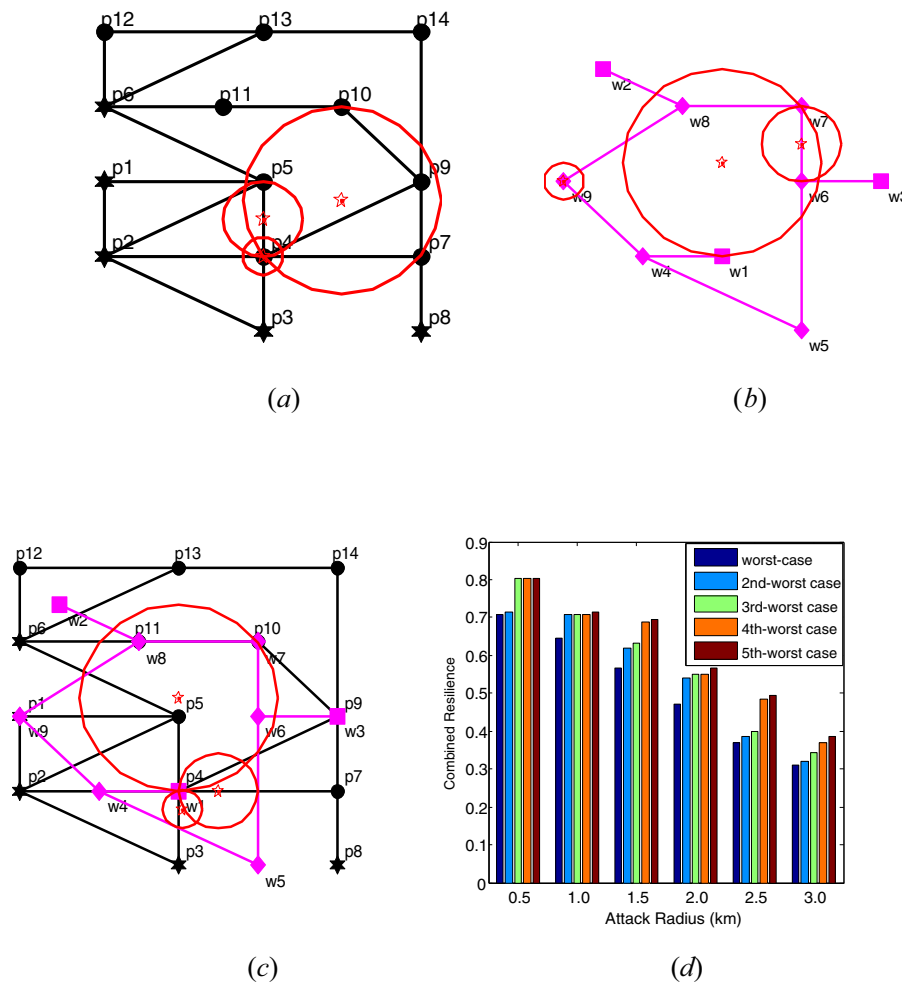


Fig. 3. Worst-case attack scenarios under several typical attack radiuses for each single system and the interdependent systems when there is no defense. (a) single power system; (b) single water system; (c) interdependent power and water systems; (d) the power and water combined resilience associated with the worst-case attack, the second-worst through the fifth-worst attacks.

Third, Fig. 5(a) shows the combined power and water resilience as a function of the attack radius under different D_B . From the figure, at small $r_A < 1.0$ kilometers, there are only a few components in the maximal component set, and then the combined resilience fluctuates a little. Also, low defense budget can bring significant resilience enhancement. When $D_B=3$ units, the resilience under $r_A < 1.0$ kilometers can be enhanced significantly from 0.715 to 0.802. When r_A becomes 1.0 kilometers, the worst-case SLA can cover more components and then make a significant resilience decrease (from 0.709 to 0.646) under no defense, but this can be effectively mitigated by small D_B . When D_B increases, the resilience curve becomes less and less steep, reflecting the effectiveness of the defense.

As the interdependencies can cause cascading failure across multiple CISs and aggravate the damage consequences, this paper next studies how single system resilience under the worst SLA will be affected between the cases of considering and not considering interdependencies. The results are shown in Fig. 5(b) for single power system and in Fig. 5(c) for single water system. From the figures, when there is no defense, the interdependencies do not affect the power resilience, but the water resilience is largely affected at most of r_A , with the resilience difference larger than 0.19. When the defense budget increases, there is no difference for the power resilience at most of r_A . This is because there is only one power generator node p1 that needs water to operate, and its failure does not affect the power supply too much (the total power

demand is 259 megawatts, but the total power supply is still 440 megawatts even under the failure of the power node p1). For the water system, with the increase of defense budget, the resilience difference keeps still large at some r_A because many water nodes depend on the electricity for operation, and the failure of the interdependent power nodes will cause the failure of the corresponding water nodes.

4.2. Importance of interdependencies

The studies in Section 4.1 are all based on the assumption that the attacker and the defender both recognize the interdependencies between multiple CISs and have the same objective function. But this assumption may not hold true in practice. The attacker may select to attack individual system by minimizing single system resilience, or the defender may distribute the defense budget to each CIS and each system protects itself individually without considering the interdependencies. This subsection will study these two assumptions to analyze the importance of interdependencies from both the defender's and the attacker's perspectives.

Assumption 1. The defender protects multiple CISs with the consideration of interdependencies to maximize the combined resilience, but the attacker only attacks single CIS to minimize single system resilience. In this case, it can be expected that

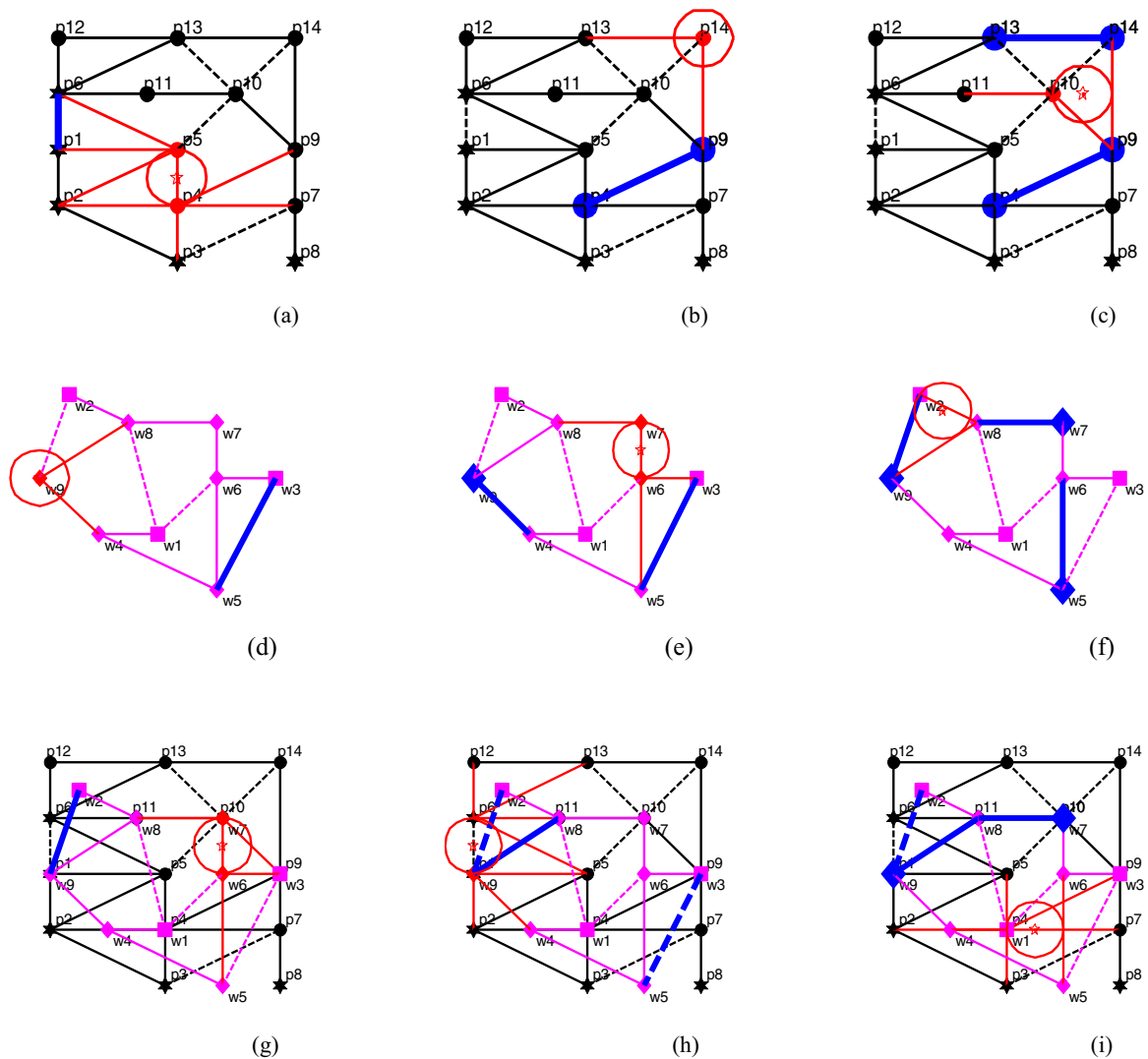


Fig. 4. The optimum defense strategy and the worst attack for each single system and the interdependent systems under different defense budgets, $r_A=1.0$ kilometers (a) single power system, $D_B=2$; (b) single power system, $D_B=5$; (c) single power system, $D_B=11$; (d) single water system, $D_B=2$; (e) single water system, $D_B=5$; (f) single water system, $D_B=11$; (g) interdependent systems, $D_B=2$; (h) interdependent systems, $D_B=5$; (i) interdependent systems, $D_B=11$.

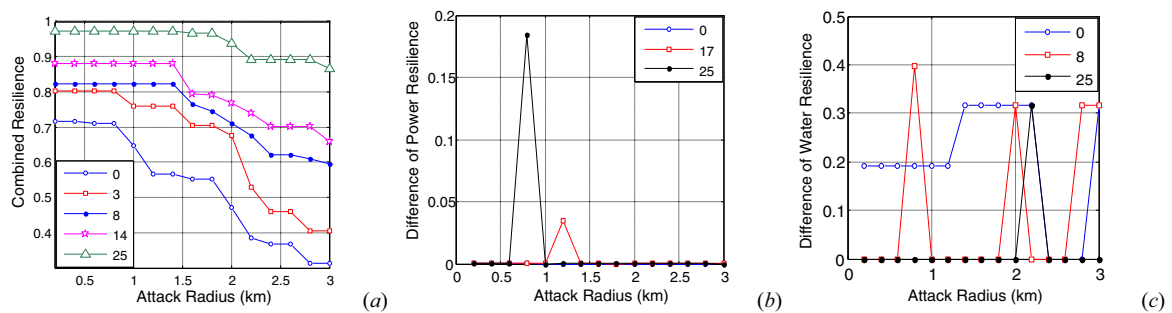


Fig. 5. (a) The combined power and water resilience curve as a function of the attack radius under different defense budgets: 0, 3, 8, 14, 25, respectively. (b) Difference of power resilience between the cases of considering and not considering interdependencies. (c) Difference of water resilience between the cases of considering and not considering interdependencies.

the combined resilience under the disruptive strategy adopted by the attacker will be larger than that for the worst SLA with the consideration of the interdependencies. Fig. 6(a) shows the combined resilience curves as a function of attack radius when the attacker makes the attack strategy, respectively, by considering the interdependencies and by focusing on each single system under no defense. From the figure, when $r_A=1.0$ kilometers, if

attacking the interdependent systems, the combined resilience is 0.64; if attacking single power system, the combined resilience is 0.79; if attacking single water system, the combined resilience is 0.76. The difference for the combined resilience in the two cases can reflect the importance of interdependencies from the attacker's perspective. Note that when the attack radius increases, the combined resilience difference decreases significantly, this is

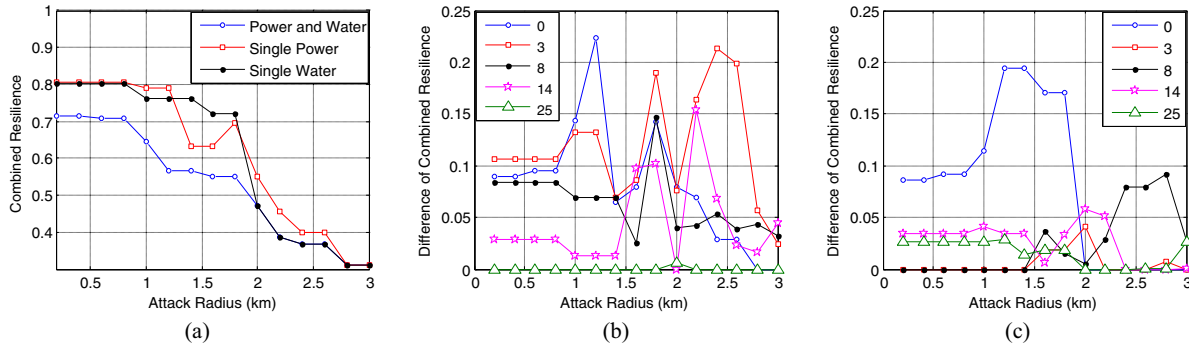


Fig. 6. (a) The combined resilience curves as a function of the attack radius under the worst attack for single power system, single water system and interdependent systems under no defense. (b) The combined resilience difference as a function of the attack radius under the worst SLA for single power system and the worst SLA for interdependent systems when the defense budget has different values: 0, 3, 8, 14, 25. (c) The combined resilience difference as a function of the attack radius under the worst SLA for single water system and the worst SLA for interdependent systems when the defense budget has different values: 0, 3, 8, 14, 25.

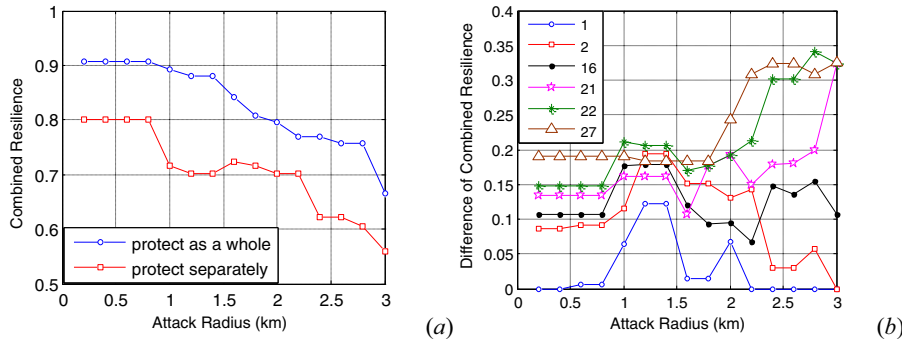


Fig. 7. (a) The combined resilience curves as a function of the attack radius, respectively, under the case of protecting multiple CISs as a whole and the case of protecting single system separately, $D_B=16$; (b) The combined resilience difference as a function of the attack radius between the case of protecting multiple CISs as a whole and the case of protecting single system separately when the defense budget has different values 1, 2, 16, 21, 22, 27.

because large attack radius causes extensive component failures and then considering or not considering interdependencies do not affect the system performance too much. Fig. 6(b) and (c) shows the combined resilience difference for each single system under different D_B . It can be found that the combined resilience difference fluctuates with the attack radius for each system, but with a trend that large defense budget makes the difference decrease close to zero as many vulnerable components would be protected.

Assumption 2. The defender distributes the defense budget evenly to each system, and each system protects itself separately without considering interdependencies, while the attacker disrupts multiple systems with the consideration of interdependencies. As expected, the combined resilience in this case will decrease compared with the case of protecting multiple systems as a whole. Fig. 7(a) shows the combined resilience curves as a function of the attack radius under these two cases when the defense budget $D_B=16$. From the figure, when $r_A=1.0$ kilometers, if protecting multiple CISs as a whole, the combined resilience is 0.8931; if protecting each system separately, the combined resilience is 0.7155, with a decrease of 0.1776. The combined resilience difference between the cases of considering multiple systems as a whole and protecting each system separately can reflect the importance of interdependencies from the defender's perspective. Fig. 7(b) shows the combined resilience difference between the two cases under different D_B . From the figure, when D_B is relatively small, the importance of interdependencies from the defender's perspective is relatively small; when D_B increases, the importance of interdependencies becomes more significant. Particularly, when D_B is 27 units and the attack radius is 1.0 kilometers, the combined resilience decreases from 0.9853 to 0.7948, with a difference of 0.1905. When

$r_A=3.0$ kilometers, the combined resilience decreases from 0.9305 to 0.6041, with a difference of 0.3264. These results indicate the importance of protecting multiple CISs as a whole under high defense budget against the spatially localized attacks.

5. Conclusions

This paper defines the set of all hazards that could cause direct localized damage or interruption as spatially localized attacks (SLAs). As SLAs can be triggered in various ways in practice, instead of defining and analyzing each type of the SLAs-related hazards, this paper makes a worst-case analysis and proposes a tri-level defender-attacker-defender model and its solution technique to support resilience optimization of interdependent CISs under the worst SLA. The interdependent power and water systems have been used to demonstrate the proposed approach. Main results include (1) the best defense strategy in the case of considering interdependencies is totally different from those for single systems; (2) low defense budget can bring significant resilience enhancement for interdependent systems against the worst-case SLA; (3) under large attack radius or high defense budget, considering the interdependencies becomes less important for the attacker; (4) under high defense budget, considering the interdependencies becomes much more important for the defender.

Numerous issues remain for future work, and they include: (1) this paper assumes there are sufficient repair resources, and then the resilience is simply quantified as system robustness. In the case of insufficient repair resources, system resilience should be measured as a combination of the robustness and the rapidity by which the system recovers after a disruption, extending the proposed approach to additionally model the post-disaster restoration

process is a direction for future research (Ouyang & Fang, 2017; Ouyang et al., 2012). (2) This paper uses a network flow model to simulate the operation of each CIS. In this case, more components' failures can cause less resilience, which might not hold true for other types of flow models, such as direct current power flow model. Extending the proposed approach to consider the case that more component failures might cause large resilience is the second direction for future work. (3) This paper assumes there is only single spatially localized attack, which may not hold true if some terrorists design the attack. Considering multiple SLAs is the third direction for future research. (4) This paper assumes that unprotected components will definitely fail if they are located within or cross the attack area. In some cases, unprotected components may fail with different probabilities, depending on their distances from the attack center (Wu, Tang, & Wu, 2016). Considering different failure probabilities for components within the attack area is the fourth direction for future research. (5) The proposed algorithm is based on iteratively solving the sub-problem and the master problem. The sub-problem needs to first identify all MCSs and then solve a MIP model for each MCS, while the master problem is also a MIP model. The computation cost for identifying all maximal component sets is $O(n^2)$ and can be significantly reduced by using parallel computing. But for the MIP model, when its solution is beyond the capacity of CPLEX, some heuristic algorithms, such as genetic algorithms, can be used as an alternative. The solution quality and efficiency of these heuristic algorithms need to be further studied and checked, which can be regarded as the fifth direction for future research.

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